

EXPERIMENT 10

(A)

Design a robust PostgreSQL transaction system for the students table where multiple student records are inserted in a single transaction. If any insert fails due to invalid data, only that insert should be rolled back while preserving the previous successful inserts using savepoints.

The system should provide clear messages for both successful and failed insertions, ensuring data integrity and controlled error handling.

CODE:

```
CREATE TABLE students (  
    id SERIAL PRIMARY KEY,  
    name VARCHAR(50) UNIQUE,  
    age INT,  
    class INT  
);  
  
DO $$  
BEGIN TRANSACTION  
    -- Start a transaction  
    BEGIN  
        -- Insert multiple students  
        INSERT INTO students(name, age, class) VALUES ('Anisha',16,8);  
        INSERT INTO students(name, age, class) VALUES ('Neha',17,8);  
        INSERT INTO students(name, age, class) VALUES ('Mayank',19,9);
```

-- If all succeed

RAISE NOTICE ' Transaction Successfully Done';

EXCEPTION WHEN OTHERS THEN

-- If any insert fails

RAISE NOTICE 'Transaction Failed..! Rolling back changes.';

RAISE; -- this will rollback the entire transaction

END;

END;

\$\$;

SELECT * FROM students;

	id [PK] integer	name character varying (50)	age integer	class integer
1	1	Anisha	16	8
2	2	Neha	17	8
3	3	Mayank	19	9